Dr. Tushar Suhasaria

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Birth: 28.02.1988, Berlin. Nationality: Indian

Education

10. 2011— 02. 2018	Ph.D. (Chemistry) (*Magna cum Laude*) Westfälische Wilhelms Universität Münster (WWU), Germany Thesis title: "Investigations of the thermal and non-thermal processing of astrophysically relevant pure and doped ices on interstellar grain analogues". Supervisor: Prof. Helmut Zacharias.
2009—2011	M.Sc. (Chemistry) (CGPA 8.92/10) Indian Institute of Technology (IIT) Roorkee, India Thesis title: "Removal of Vanadium from aqueous solutions using Red Mud- an aluminum industry waste as adsorbent". Supervisor: Prof. Dr. Vinod Kumar Gupta.
2006—2009	B.Sc. (Hons. Chemistry) (Distinction, 70%) Bankura Christian College, University of Burdwan, India Subjects: Chemistry (Major), Physics and Mathematics (Minor).

Scholarships/ Awards

06.2016	*COST Action Research and Travel Fellow* (CM1401) for Short-Term Scientific Missions (STSM), INAF Catania Astrophysical Observatory, Italy.
02.2014— 10.2014	*Doctoral Research Scholarship* Awarded for conducting research at the Graduate School of Chemistry, WWU Münster, Germany.
10. 2011— 01. 2014	*ITN Marie Curie Fellowship* for Early Stage Research for European Union funded project LASSIE (Laboratory Astrochemical Surface Science in Europe).
2011	*JRF (Junior Research Fellow)* and NET (National Eligibility Test for College and University Teaching), awarded for starting doctoral research and teaching as an assistant professor in any higher education institute of India, Council of Scientific and Educational Research, India.
2009	*Professor A.P Kundu Memorial Prize* for securing highest marks in Organic Chemistry, B.Sc. (Hons.), University of Burdwan, India.

Research Experience

03.2021—	*Post-doctoral Researcher,* Max Planck Institute für Astronomie, Heidelberg, Germany Nature of Work: Atom bombardment, Prebiotic chemistry, Photochemistry of ices, Infrared spectroscopy, Building of state-of-the-art Ultra high vacuum setup for in-situ Laser desorption-ionization experiment
10. 2018— 09. 2020	*Post-doctoral Researcher,* INAF Capodimonte Astronomical Observatory, Naples, Italy.

	Nature of Work: H atom and UV photon irradiation experiments, transmission infrared spectroscopy, cryogenics, chemical handling, secondary electron microscopy.
10. 2014— 02. 2018	*Doctoral Research Assistant,* Physics Institute, WWU Münster, Germany Nature of Work: Temperature Programmed Desorption (TPD), Reflection Absorption Infrared Spectroscopy (RAIRS), AFM, cryogenics, XUV photon irradiation experiments, UHV, TOF-SIMS.
04.2013— 06.2013	*Research Assistant* (Academic Secondment), INAF Catania Astrophysical Observatory. Nature of Work: Ion irradiation experiment, transmission infrared spectroscopy and cryogenics.

Training/ Internship

08.2015— 09.2015	Research assistant, DESY-Free Electron Laser (FLASH), Hamburg, Germany. Nature of Work: Photochemistry induced by XUV radiation, time of flight mass spectrometry, cryogenics, resonance enhanced multi-photon ionization, lasers.
05.2010— 06.2010	Summer Intern, Indian Institute of Petroleum, Dehradun, India Nature of Work: X-ray diffraction, FTIR, organic synthesis, catalysis.

Teaching/ Mentoring

WS2024	Supervision of Bachelor Thesis: Leuschner, Vanessa “Temperature Programmed Desorption Experiments on interstellar Dust Grain Analogues”.
SS2024	Supervision of research Internship: Eckhardt, Frederick. Ex-situ analysis of formation of prebiotic molecules in interstellar ice analogues via mass spectrometry
SS2024	Supervision of Master Project: Abdo, Eddy. Solid-state processing of interstellar ice analogs
SS2024	Supervision of research Internship: Efthymiou, Zoi. Temperature Programmed Desorption Experiments on amorphous carbon and silicate grain analogues
SS2023- WS2024	Supervision of Master Thesis: Wee, Soo May “Experimental Characterization of Astrophysically Relevant Molecules on Dust-Ice Interfaces”.
SS 2023	Supervision of research Internship: Paradis, Arthur. Synthesis of organic molecules under interstellar conditions.
SS 2023	Supervision of research Internship: Abdo, Eddy. Studying the efficiency of an atomic source
SS 2022	Supervision of research Internship: Baecker, Tia. Involved partly in the building up of Orbitrap-MS coupled UHV setup.
WS 2021	Supervision of research Internship: Demertzi, Dimitra. Involved in the building up of Orbitrap-MS coupled UHV setup.
WS 2021	Supervision of Bachelor Thesis: Pérez Rickert. Paula Caroline “Diffusion of

	CO_2 Molecules on the Surface of Compact Amorphous Solid Water".
SS 2021	Supervision of Summer Internship: Sohier, Orianne. "Design and Characterization of the Ultra-High Vacuum setup in Origins of Life Laboratory".
WS 2017	Supervision of Bachelor Thesis: Glienke, Marcel. "Investigation of coronene adsorbed on HOPG with infrared spectroscopy and temperature programmed desorption".
SS 2014	Supervision of Bachelor Thesis: Wettstein, Alina. "Characterisation of ammonia ice on forsterite interstellar grain mimic".
SS 2014— SS 2017	Functional Nanosystem Masters Practical Courses Topics: TPD and RAIRS.

Skills

Languages	➢ Native Language Proficiency: English, Hindi and Bengali. ➢ Elementary Proficiency: German (B1) and Italian (A2).
Management (Selected)	➢ Core Team Member of the Organizing Committee for the upcoming European Conference on Laboratory Astrophysics "ECLA 2020", Capri. ➢ Conducted Outreach Workshop for young students at the International Astronomical Union S-350 Symposium in Cambridge, UK, 2019. ➢ Convener of Administration and Finance of "COGNIZANCE 2011" at the Department of Chemistry, IIT Roorkee, India.
Communication	➢ Regularly presented research output in 10+ international conferences, symposiums and workshops.

Service

Referee for A&A, ApJ and PRL

Publications

T. Suhasaria, S M Wee, R Basalgète, S A Krasnokutski, C Jäger, G Perotti, Th Henning (2024). Infrared spectra of solid-state ethanolamine: Laboratory data in support of JWST observations. Astron. Astrophys., 691 A306.

T. Suhasaria, Th. Henning and V. Mennella (2024). Stability of solid-state formamide under Lyα irradiation. Astron. Astrophys., 683 A92.

J. He, P. C. Pérez-Rickert, **T. Suhasaria**, O. Sohier, T. Bäcker, D. Demertzi, G. Vidali, and T. K. Henning. (2023). New measurement of the diffusion of carbon dioxide on non-porous amorphous solid water. Molecular Physics: e2176181.

T. Suhasaria, V. Mennella (2022). Lyα irradiation of solid-state formamide. Astron. Astrophys., 662 A73.

T. Suhasaria, V. Mennella (2021). Catalytic Role of Refractory Interstellar Grain Analogs on H_2 Formation. Frontiers in Astronomy and Space Sciences, 8, 655883.

V. Mennella, **T. Suhasaria**, L. Hornekær, J. D. Thrower, and Giacomo Mulas (2021). Lyα Irradiation of Superhydrogenated Coronene Films: Implications for H_2 Formation. The Astrophysical Journal Letters, 908(1), p.L18.

T. Suhasaria, V. Mennella (2020). Destruction route of solid-state formamide by thermal H atoms, Astron. Astrophys., 641, A88.

V. Mennella, M. Ciarniello, A. Raponi, F. Capaccioni, G. Filacchione, **T. Suhasaria**, C. Popa, D. Kappel, L. Moroz, V. Vinogradoff, and A. Pommerol (2020). Hydroxylated amorphous silicates: a new component of 3.4 micron band in 67P/CG comet. Astrophys. Journ. Lett., 897, L37.

T. Suhasaria, J. D. Thrower, R. Frigge, S. Roling, M. Bertin, X. Michaut, J.-H. Fillion, H. Zacharias (2018). XUV photodesorption of carbon clusters ions and ionic photofragments from a mixed methane-water ice. Phys. Chem. Chem. Phys., 20, 7457.

T. Suhasaria, G. A. Baratta, S. Ioppolo, H. Zacharias, M. E. Palumbo (2017). Solid CO_2 in quiescent dense molecular clouds- comparison between Spitzer and laboratory spectra. Astron. Astrophys., 608, A12.

T. Suhasaria, J. D. Thrower, H. Zacharias (2017). Thermal desorption of astrophysically relevant molecules from forsterite(010). Mon. Notices Royal Astron. Soc., 472, 389.

T. Suhasaria, J. D. Thrower, H. Zacharias (2015). Thermal desorption of ammonia from crystalline forsterite surfaces. Mon. Notices Royal Astron. Soc., 454, 3317.